

Designing Management Controls for Skill Development: Can Technology Substitute for Supervisor Engagement?

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Abstract: Firms increasingly rely on technology to support employee skill development, raising the question of whether such technology complements or substitutes for the supervisor's role. We study this question in a nine-month field experiment with a large European service firm, randomly assigning 187 teams to a control condition, a complement condition that makes development conversations salient and prompts supervisors to hold them, or a substitute condition that lets supervisors schedule training directly without a development meeting. The complement condition increases trainings, development interactions, employee knowledge, job satisfaction, perceived supervisor support, retention, and performance on the most skill-intensive tasks. The substitute condition raises employee knowledge but moves none of the broader outcomes. Technology could substitute for the supervisor in producing the measurable skill gain, but not in producing the relational and operational benefits that flow through supervisor engagement. Our findings inform the design of management controls for skill development.

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1. INTRODUCTION

In this study, we examine how to support supervisors' engagement in developing the skills of their subordinates, without necessarily changing the firm's formal incentive system. Firms invest substantial resources in developing the skills of their employees (Freifeld 2023; SHRM 2026). While prior accounting research has primarily examined how to motivate employee effort on current tasks, recent work has started to examine how firms manage the development of employee skills (Künneke 2024; Arnold, Shi, Tafkov, and Voermans 2025; Grabe and Sliwka 2025). In many firms, the supervisor is the central actor in this process. Supervisors decide when subordinates participate in training, identify their development needs, and hold development conversations. Their engagement with these activities affects subordinate outcomes (Künneke 2024; Grabe 2025; Grabe and Sliwka 2025). However, it is difficult for firms to ensure such supervisor engagement because performance in these activities is rarely measured and supervisors face competing demands on their time. For example, in our setting, a large multinational service firm, the firm requires supervisors to hold a development conversation with each subordinate every quarter. Yet supervisors hold, on average, approximately one such conversation per year. This gap reflects a tension between at least two tasks that compete for the supervisor's attention. The first is the operational task of running the team, which is measured and monitored daily. The second is subordinate development, which is less frequently monitored, leading to classical multitasking tradeoffs (Holmstrom and Milgrom 1991; Feltham and Xie 1994).

We focus on two distinct approaches a firm can take to address this problem through its management control system. The first is to increase the supervisor's engagement with the development task, prompting the supervisor to invest more in an activity they would otherwise economize on. The second is to reduce the development process's reliance on the supervisor,

allowing the firm's information system to take a larger role in converting identified development needs into action. The two approaches differ in the role they assign to the supervisor relative to the technology. The first treats a technology feature as a complement to supervisor engagement, providing information that makes the supervisor's involvement more effective. The second treats it as a substitute, allowing technology to carry more of the development process so that outcomes can be produced with less supervisor involvement. Whether a technology feature can substitute for supervisor engagement in producing development outcomes, or whether supervisor engagement contributes value that a substitute cannot reproduce, is a key question that we examine. While prior accounting work has established skill development as an important management control problem, with the supervisor at its center (Künneke 2024; Arnold et al. 2025), field-experimental evidence on this question is limited.

We collaborate with a large European service firm and randomly assign 187 teams (3,644 service technicians and their supervisors) to one of three treatments over a period of nine months. In the control group, supervisors receive information about subordinates' training needs from the firm's digital learning platform. In the complement treatment, supervisors additionally receive information about whether they have engaged in development conversations with their subordinates in the current quarter, together with prompts to do so. This treatment is designed to increase the supervisor's engagement with development. The substitute treatment instead allows supervisors to act on the platform's training recommendations directly, scheduling a training session in response to a subordinate's knowledge test without first holding a development conversation. This treatment is designed to let the platform carry more of the development process, reducing the role of the supervisor's conversation. Comparing the two treatments allows us to ask whether the platform can substitute for the supervisor's engagement in producing development outcomes. We measure outcomes from personnel records as well as employee and

customer feedback. These include the number of training sessions, knowledge test scores, the frequency of supervisor-subordinate development interactions, perceived supervisor support, job satisfaction, turnover intentions, and operational measures of team performance.

Our main results are as follows. First, the complement treatment shifts supervisor behavior. Compared to the control group, it increases the number of training sessions scheduled by approximately 15% and the frequency of supervisor-subordinate development interactions by approximately 14%. The substitute treatment does not increase the number of training sessions or the frequency of development interactions. However, both treatments significantly increase test scores, which are a key measure of skill development. Second, the two treatments differ on the broader employee outcomes the firm cares about. The complement treatment increases job satisfaction and perceived supervisor support, decreases turnover intentions, and improves operational performance, reducing completion time on the firm's most skill-intensive tasks by approximately 1.3 minutes per assignment (4% improvement). The substitute treatment produces only a smaller reduction in completion time that is spread across all tasks rather than concentrated in the skill-intensive ones, and it does not significantly affect employee attitudes. The complement treatment thus improves performance precisely where accumulated skill matters most, while the substitute treatment does not. Third, neither treatment reduces customer service quality or work quality. Directing supervisors' attention toward employee development therefore did not come at the expense of the operational tasks the firm measures, and in the case of the complement treatment it improved the most skill-intensive ones. This runs counter to the standard multitasking concern that directing attention toward an unmeasured task crowds out performance on the measured tasks. Fourth, the firm subsequently decided to roll out the complement treatment to its full workforce, providing external validation of the design choice.

We conclude that our results show a nuanced picture on our substitution question. The platform feature in the substitute treatment was able to produce the proximate output of the development process, employee knowledge, with little supervisor involvement. But it did not reproduce the broader outcomes that the complement treatment produced, all of which depend on the supervisor's engagement with the subordinate. In our setting, technology could substitute for the supervisor in delivering the measurable skill gain, but not in delivering the relational and operational benefits that flow through supervisor engagement.

Our paper makes several contributions. First, we add to recent accounting literature on the design of management controls for skill development (Künneke 2024; Arnold et al. 2025) by providing field-experimental evidence on the supervisor side of the development process. Arnold et al. (2025) study the effect of relative performance information on employee skill investment in the laboratory. Using proprietary archival data, Künneke (2024) shows that supervisor engagement with performance evaluation predicts subordinate development and career outcomes. We add field-experimental evidence on how the management control system can shift supervisor engagement with development. Our results also complement Grabe and Sliwka (2025), who show that removing skill information from supervisors harms development outcomes. Building on this evidence that the information environment matters, we examine which design choices in the firm's learning platform direct supervisor engagement with subordinate development, and whether that engagement can be substituted by the platform itself.

Second, we provide evidence on the limits of substitution between technology features and direct supervisor engagement in a management control process. Both treatments raise the proximate output of the development process, employee knowledge. Yet only the complement treatment raises the broader outcomes that depend on the supervisor's role in the development relationship: perceived supervisor support, operational performance, job satisfaction, and

retention. The substitute treatment produces the knowledge gain with little supervisor involvement but does not reproduce these broader outcomes. This pattern suggests that the supervisor's engagement produces value beyond the proximate output of the development process, and that this value is difficult to substitute through a technology feature that reduces the supervisor's role. The finding speaks to the question of which elements of a management control system can be substituted by technology and which cannot (Grabner and Moers 2013) and provides robust evidence that elements operating through direct supervisor engagement are difficult to replace.

Furthermore, for firms, the results point to a practical design principle. When an important task may be underweighted because it is hard to measure, a low-cost modification to the information system that makes the task salient to managers can shift their behavior toward it, without renegotiating incentives or reducing quality of other dimensions. At the same time, firms should be cautious about using technology to substitute for the manager in such tasks: in our setting, the feature that reduced the manager's role delivered the measurable output but appears to reduce the relational and operational benefits that came from the manager's engagement.

2. THEORETICAL BACKGROUND AND HYPOTHESES DEVELOPMENT

Skill development as a management control problem

Firms invest in the skills of their employees, and in many organizations the supervisor is the key figure in this process. A large literature documents that supervisors have substantial effects on the productivity, retention, and development of the workers they manage which ultimately benefits firm performance (Bertrand and Schoar 2003; Lazear, Shaw, and Stanton 2015; Hoffman and Tadelis 2021). Among the channels through which supervisors create value, several involve the development of subordinates. Supervisors identify development needs, assign training, coach and

mentor, and conduct performance evaluations through which development is planned and assessed (Lazear, Shaw, and Stanton 2015; Benson and Shaw 2025). In the accounting literature, recent evidence shows that the way supervisors conduct performance evaluation predicts subordinate development and career progression, and that supervisors' engagement with the evaluation process varies in ways that determines its developmental value (Künneke 2024). At the same time, development activities are difficult for the firm to observe and measure, and firms commonly mandate them through policy, such as a required periodic development conversation, without measuring whether the activity takes place or tying it to supervisor compensation. The setting we study has this structure, as described in Section 3.

This produces a multitask problem where a supervisor allocates effort across at least two tasks. The first is the operational task of running the team, on which the firm collects performance data continuously and against which it monitors and rewards supervisor effort. The second is subordinate development, which the firm requires but does not measure. When tasks differ in how precisely they can be measured, incentives and attention attach more readily to the better-measured task, distorting the agent's effort allocation away from the task whose contribution is harder to capture (Holmstrom and Milgrom 1991; Feltham and Xie 1994). The resulting distortion has been characterized in terms of the congruity of the available performance measures (Feltham and Xie 1994; Datar, Kulp, and Lambert 2001) and the difficulty of aggregating performance across heterogeneous tasks (Şabac and Yoo 2018). Because operational performance is measured continuously while development is not, supervisors have reason to underinvest in development even when the firm values it.

Supervisors do not always engage with the management control activities the firm assigns to them, and the development conversation is one such activity. Conducting a meaningful evaluation requires the supervisor to gather information and to differentiate among subordinates,

both of which are costly. When these costs are high, supervisors economize on evaluation effort, producing compressed and lenient ratings that carry little developmental information (Moers 2005; Bol 2011). Bol (2011) shows that higher information-gathering costs increase these biases and that the biases affect not only current ratings but future employee incentives. Künneke (2024) shows that supervisors who differentiate more in evaluation produce better subordinate development and career outcomes. Collectively, existing evidence indicates that there is substantial heterogeneity in supervisor engagement in employee development, even though that engagement is valuable to the subordinate and firm.

Attention-directing mechanisms

A central insight of the management control literature is that the firm does not necessarily need to change the explicit incentive contract to influence the supervisor's effort allocation. The elements of a management control system are interdependent, so the firm can change behavior by altering how those elements relate to one another rather than by changing incentives (Grabner and Moers 2013). The implicit weight a supervisor places on a task can be shifted by features of the management control system that direct attention toward that task. Casas-Arce, Lourenço, and Martínez-Jerez (2017) show that the frequency and detail of nonfinancial performance feedback shift the behavior of professionals in the field. Relatedly, Manthei, Sliwka, and Vogelsang (2023) show that introducing regular conversations in which managers report their activities toward a profit objective to their supervisor raises performance on that objective substantially, and at low cost, without changes to financial incentives. Friebe, Heinz, and Zubanov (2022) document that a one-page letter from the CEO of a retail chain urging store managers to address employee turnover shifts middle-manager effort toward that objective, again without any change in the formal incentive contract, although the effect fades and reappears only when managers are

reminded. Sliwka (2007) shows theoretically that the design of a firm's control system is not informationally neutral, and the choices a firm makes can reveal information to employees and guide their response.

Hypothesis development

The previous section establishes that supervisors underinvest in subordinate development because development is an unmeasured task that competes for attention with the operational task the firm monitors continuously, and that the implicit weight a supervisor places on such a task can be shifted by features of the management control system without changing the formal incentive contract. We develop predictions for two such features, which correspond to two treatments in our experiment. The first complements the supervisor's role by directing attention to the development task. The second substitutes for part of the supervisor's role by removing a step that stands between an identified development need and a scheduled training.

The development process in our setting combines several elements. The knowledge test identifies an employee's skill gaps, the employee submits ideas for their own development, and the supervisor decides what training the employee attends. In the firm's intended process, these elements come together in the development meeting, where the supervisor and employee discuss the test results and the employee's ideas and decide on training. The meeting is thus the step that converts identified needs into training decisions. The firm can use its learning platform to change the role of this step. It can reinforce the meeting, prompting the supervisor to hold it and to use the employee's ideas as its basis, so that the supervisor's engagement in the meeting becomes more central to the development decision. Alternatively, it can allow the supervisor to act on the employee's submitted ideas and test results directly, scheduling a training without holding the meeting, so that the meeting is no longer a necessary step between an identified need and the

training that addresses it. Whether the supervisor's engagement in the meeting can be removed in this way without loss, or whether the meeting contributes value to development outcomes beyond the training decision it produces, is the question we examine.

Complementing supervisor engagement by directing attention to development

When a valued activity carries up-front costs and delayed, hard-to-measure benefits, agents tend to defer it, and limited attention is a primary reason. The activity drops out of the agent's consideration set in favor of more immediate and more salient demands (Calzolari and Nardotto 2017). A direct implication is that returning the activity to the agent's attention shifts behavior toward it, helping to align the agent's present actions with the future value the activity produces (Milkman et al., 2021; Barron, Damgaard, and Gravert 2022). Calzolari and Nardotto (2017) show in a field experiment that periodic reminders substantially increase engagement in an investment activity with up-front costs and delayed benefits, with the largest effects among agents who would otherwise act least often. In the case of employee management, Friebe et al. (2022) study the effectiveness of a randomly distributed letter by the CEO of a retail chain, urging store managers to “do what they can” to reduce employee turnover. As a response, store managers who received the letter increased their focus on HR-related behavior, reducing turnover by 20%. In our setting, assigning a training session is exactly the kind of deferrable action this mechanism predicts supervisors will postpone. Assigning training carries an immediate cost in the supervisor's time and yields a benefit to the employee's skills that is delayed and hard to measure, while the operational task competing for the supervisor's attention is monitored daily. Supervisors in our firm report postponing or canceling trainings for precisely this reason, holding far fewer development conversations and assigning fewer training sessions than the firm intends. A treatment that makes development salient and prompts the supervisor to act is expected to

counteract this postponement, returning the deferred training decisions to the supervisor's attention and raising the rate at which they are made. Our first treatment, the contact support treatment, is designed to do this: it displays whether development meetings have taken place and prompts the supervisor to hold those that are outstanding. Because this treatment works by strengthening the supervisor's own engagement with the development task, we also refer to it as the complement treatment. We predict that it increases the number of training sessions the supervisor assigns.

H1a. The contact support treatment increases the number of trainings assigned by the supervisor.

The attention-directing mechanism also bears on the development meeting itself. The development meeting is the firm's designated occasion for assessing a subordinate's skills and planning their development, and it is a form of subjective evaluation that supervisors find costly and tend to economize on (Moers 2005; Bol 2011; Künneke 2024). Manthei, Sliwka, and Vogelsang (2023) show that prompting regular structured conversations about an objective raises engagement with that objective in the field. Because the contact support treatment directs the supervisor's attention specifically to whether development meetings have taken place, and prompts the supervisor to hold those that are outstanding, we expect it to increase the frequency of development interactions between supervisor and subordinate.

H1b. The contact support treatment increases the frequency of supervisor-subordinate development interactions.

Trainings and development meetings are the proximate inputs to a subordinate's skill development. Employer-provided training raises worker skills and productivity (Konings and Vanormelingen 2015), and the development meeting is the firm's mechanism for matching training to the subordinate's needs and for reinforcing the subordinate's engagement with development (Künneke 2024). If the contact support treatment increases both the number of trainings and the frequency of development meetings, as predicted in H1a and H1b, it should raise the subordinate's measured knowledge.

H1c. The contact support treatment increases employee knowledge as measured in the knowledge test.

Substituting for supervisor engagement by removing the coordination step

Attention is not the only barrier to action. Even when an agent attends to a valued activity, a required intermediate step can stand between the agent and completion, and removing that step raises the rate at which the action is taken. This is among the most robust findings in the literature on choice architecture. Choi, Laibson, and Madrian (2004) show that reducing the steps required to enroll in a savings plan raises participation substantially, and Beshears, Choi, Laibson, and Madrian (2013) show that collapsing a multistep decision into a single step raises take-up by 10 to 20 percentage points. In the development process we study, acting on a subordinate's identified development need ordinarily requires the supervisor first to coordinate with the subordinate through a development meeting before a training is scheduled, so the meeting represents an intermediate step between an identified need and the training that addresses it. Our second treatment, the scheduling support treatment, is designed to remove this step. It allows the

supervisor to act on a subordinate's identified training needs directly through the platform, without first holding a development meeting. Because this treatment lets the platform carry the development decision in place of the supervisor's coordination with the subordinate, we also refer to it as the substitute treatment. We argue that lowering expected costs associated with scheduling employee training increases the likelihood of supervisors using the platform to schedule training sessions for their employees.

H2a. The scheduling support treatment increases the number of trainings assigned by the supervisor.

The scheduling support treatment allows the supervisor to act on identified needs without the development meeting, and presents this as the intended way to use the feature. The treatment therefore substitutes direct action on the platform for the supervisor-subordinate coordination that the meeting usually provides. We expect supervisors in this treatment to hold fewer development meetings with their subordinates than in the control group.

H2b. The scheduling support treatment decreases the frequency of supervisor-subordinate development interactions.

The substitute treatment is predicted to reduce the frequency of development meetings (H2b), which raises the question of whether reducing supervisor contact might in turn harm employee development. We argue that, for the knowledge employees acquire, it need not. In our setting, the development meeting serves as a coordination step in which the supervisor and employee turn the employee's surfaced development needs and knowledge-test results into a training decision. The

substitute treatment does not remove the inputs that drive learning, the employee's identified needs and the test results, but allows the supervisor to act on them without first holding the meeting. Evidence from a field settings suggests that reducing the supervisor's intermediation in this way can still shift the proximate outcome: Bernstein and Li (2026) show that giving front-line service employees direct access to information that had previously reached them only through their supervisor changes employee behavior on the measured margin. Because the training that produces knowledge still takes place, and continues to respond to the employee's identified needs, a reduction in the frequency of the meeting need not sever the path from identified needs to skill acquisition. As in the complement treatment, trainings are the proximate input to a subordinate's skill development (Konings and Vanormelingen 2015). If the substitute treatment increases the number of training sessions, as predicted in H2a, the resulting trainings should raise the subordinate's measured knowledge, even as development meetings become less frequent.

H2c. The scheduling support treatment increases employee knowledge as measured in the knowledge test.

The open question: complement or substitute?

The six hypotheses above predict that each treatment shifts behavior and raises knowledge, but they do not, on their own, answer the question that motivates the study: whether technology can substitute for the supervisor's engagement in producing development outcomes, or whether that engagement contributes value a substitute cannot reproduce. The two treatments embody opposing answers, and ex ante either appears defensible.

On one view, the supervisor's engagement is primarily a means of getting the right training to the right employee. The diagnostic content of the development process, the identification of what an employee needs, originates largely with the employee, who completes the knowledge test and surfaces development needs. If this diagnostic content is what drives subsequent skill gains, then a treatment that lets the employee's identified needs reach action directly should reproduce the outcomes of supervisor engagement, and may do so at lower cost, because the development meeting adds a coordination step without adding diagnostic value. On this view, the substitute treatment should achieve outcomes comparable to the complement treatment, and substitution is feasible.

On the opposing view, the supervisor's engagement produces value beyond the diagnosis it generates. The development meeting is not only an occasion to identify training needs; it is also a direct interaction in which the employee experiences the supervisor's attention and support, and through which the supervisor learns about and invests in the employee. If these features of the interaction matter for outcomes such as perceived supervisor support, job satisfaction, and retention, then a treatment that removes the meeting will raise the proximate output of the development process, employee knowledge, without reproducing the broader outcomes that depend on the interaction itself. On this view, substitution is incomplete: technology can carry the diagnostic function but not the relational one.

Distinguishing these views requires comparing the two treatments not only on the proximate outcomes covered by our hypotheses but also on the broader outcomes the firm values. We therefore examine, the effects of the two treatments on perceived supervisor support, job satisfaction, turnover intentions, and operational performance, and whether the two treatments differ on these dimensions.

3. METHOD

Experimental Setting

We run the experiment in a large service organization with more than 4,500 service technicians, whose main task it is to provide technical services to customers, such as installing or maintaining products and services of the company. As most assignments require manual work on site, employees spend most of their time at the client's facility. While employees receive their daily work schedule from the regional dispatcher, the role of the supervisor is focused on managing the workload of the team and ensuring that employees have sufficient technical knowledge to complete their daily assignments in a given time frame. The employee development process of the firm and the targeted steps of the experimental treatments are shown in Figure 1.

Given the range of services the company offers, all employees need to take part in regular knowledge assessments on key skills they possess. The assessments are conducted through multiple choice tests on an online platform, which can be accessed via tablet, phone, or laptop. As a large part of the workforce is supposed to have sufficient skill levels in certain key knowledge domains, most employees are subject to the same assessments. As the employees take part in the knowledge assessments multiple times a year, supervisors are instructed to conduct quarterly meetings to assess the current skill level of an employee, to identify knowledge gaps and to sign up the employee for a workshop or a webinar to improve or maintain their skill level. In these meetings, employees are encouraged to bring their test results to the meeting and use them as a basis for their conversation. While the firm's development guidelines posit that the development meetings should take place every quarter, most employees report that they had one development meeting with their supervisor per year. During our conversations with the firm, two reasons for this gap have emerged: First, the learning process in the firm is complicated and supervisors need to combine information from multiple sources to make training decisions for their employees.

Second, while employee skill development is important for the firm, managing the day-to-day business is more appreciated by the company's management. As a result, many managers feel inclined to cancel or postpone trainings for their employees. To simplify the learning process and ensure the long-term productivity of employees, we designed two treatments on the company's internal learning platform to underline the value of employee skill development and to support the training process in different ways.

Insert Figure 1 about here

Experimental Design

We randomly assigned 187 teams (187 supervisors and 3,644 employees) to one of three experimental groups. In both treatment groups and the control group, employees are encouraged to develop ideas for learning activities and send these ideas to their supervisors. To ensure that employees and supervisors in each treatment group have the same level of information about the treatments, the firm also informed the employees about the changes affecting their supervisors as part of the rollout of another function. Below are the messages sent to the supervisors regarding the changes on the learning platform. All messages have been edited so that the identity of the company is not revealed. The text for the control group is as follows:

Starting today, your employees can submit ideas for their development after completing the knowledge test. The ideas are available for you on the learning platform, and you can use them to make decisions about the development of your employees.

In the complement treatment, supervisors receive new features on the learning platform that make the development meetings more salient. In an email towards the end of every quarter, the organization urges them to schedule the development meetings they are obliged to hold with

every employee in the team to make decisions about their skill development. For the introduction of the new feature, they receive the following text in addition to the control message:

Make sure that you use the ideas of your employees as a basis for the development meetings. To keep track of all development meetings, you can monitor all your conversations in an overview on the learning platform. You will receive an automated email twice a quarter so that you do not forget any conversations.

In the substitute treatment, we add a new function to the learning tool to facilitate the process of scheduling trainings when using the learning tool. More specifically, supervisors can view the employee's suggestions and immediately schedule a training through the companies digital learning tool. That is, supervisors are advised to take immediate action based on the suggestions from the platform and the employee, regardless of whether the development meeting has already taken place or not. In addition to the control message, they receive the following text for the introduction of the feature:

After viewing the suggestions from your employee, you can directly switch to the training tool or the time management tool to schedule a training for your employee. This new function should help you to close skill gaps immediately after the knowledge test and without having to wait for the development meeting.

Experimental Procedure

Treatment assignment was performed four weeks prior to the experiment. We use stratified randomization on the level of a region in the company, team size and participation and outcomes in prior knowledge tests to reduce the influence regional differences and team characteristics. The experiment started in July 2022 and ran until the end of March 2023. Both control and treatment groups receive information about upcoming changes in the learning tool via email. In the

following weeks, all employees and supervisors were invited to join two virtual Q&A sessions concerning their new features on the learning platform. Q&A sessions were separated between supervisor and employees as well as the respective treatment group to avoid spillover effects between treatments. The post-experimental survey was sent out on behalf of the CEO of the company via email. All participants who fully complete the survey receive a monetary bonus provided by the university via bank transfer. Unless stated otherwise, we use all data available from the beginning of 2022 until the end of the post-experiment in March 2023. For the data analysis, we exclude all employees who changed teams up until March 2023.

Measures

Training Intensity. To assess the effect of the treatments on development investment, our main pre-registered outcome is the number of training sessions a team receives in the treatment period, which is obtained from a data set of all registered training sessions per team during the treatment period.

Employee Knowledge. We measure the current level of employee knowledge through a standardized knowledge test that almost all employees have to take part in as part of their learning journey. For each test, employees receive a score which is defined by the ratio of correctly answered multiple-choice questions and values range from 0 to 100 percent. As employees can take part in more than one test, we standardize each test and compute an average across all tests for each employee. We differentiate between tests conducted *during* the treatment period (all tests scheduled between July 1st 2022 and March 31st 2023) and *after* the treatment period (all test scheduled between April 1st 2023 and December 31st 2023).

Job Performance. Job performance is measured as the weekly average time a team needs to complete a job assignment. We run the main analyses for the most important tasks which are customer service, troubleshooting, documentation and quality control. Customer service ($M =$

6.27 min) includes all tasks associated with customer interaction prior and during the appointment. For most clients, employees are required to call the customer to understand their infrastructure and their current issue. This short conversation often serves as a precursor to on-site troubleshooting. Troubleshooting ($M = 43.29$ min) is the core of a technician's job and involves all tasks associated with problem-solving at the client site. Troubleshooting is the most time-intensive task where technical knowledge and training are most likely to play a significant role. Quality control ($M = 10.22$ min) follows troubleshooting and involves assessing the client's infrastructure, typically after resolving the issue. If any further problems remain, this serves as documentation for a follow-up visit. Finally, documentation involves all tasks associated with paperwork or data entry ($M = 4.48$ min).

Supervisor Attention. To capture the supervisor's attention to employee development, we assess supervisor behavior both using self-reports by the employees as well as calendar data from the teams work schedule. The self-report measure concerns the perceived frequency of supervisor-employee interactions (Antonakis and Atwater, 2017), captured by the amount of development meetings supervisors have conducted with their employees using the following question from the post-experimental survey: "*Thinking back to the last 6 months: How often did you have an appointment with your supervisor to talk about your skill development?*" (values ranging from 0 to "4 or more"). As an objective measure of the supervisor's investment in employee development, we obtain data from the work scheduling tool of the firm, which captures three aspects of employee development. The schedule captures time blocked for formal and informal meetings where the former captures mandatory structured performance reviews and the latter concerns ad-hoc 1:1 and team meetings to discuss current issues.

Employee Perceptions. To assess the downstream effectiveness of the treatment on broader employee development outcomes, we use multiple measures from the post-experimental

survey. All items are answered on a 7-point scale ranging from 1 (= *strongly disagree*) to 7 (= *strongly agree*), unless indicated otherwise. Whenever necessary, items were translated from English to German using back-to-back translation. We capture job satisfaction (“*All things considered, how satisfied are you with your job at the moment?*”) using a single-item measure from the Linked Personnel Panel (Kampkoetter et al., 2016), voluntary turnover intention (“*I frequently think about quitting my job.*”) again using a single-item measure from the Linked Personnel Panel (Kampkoetter et al., 2016), and perceived supervisor support, as defined in Sonntag, Schaper, and Friebe (2005) using a four-item scale from Decius, Schaper, and Seifert (2021), which has been designed to specifically for the use with blue-collar workers. A sample item is “*My supervisor is interested in what I am currently learning.*” Cronbach’s alpha of the scale is 0.92.

4. RESULTS

Summary Statistics

Table 1 summarizes descriptive statistics for all relevant outcome variables on the team level for the full sample and each treatment group. In right most column we show results from F-tests to demonstrate that randomization was balanced across treatments prior to the treatment. Descriptive statistics regarding age and tenure show that external turnover in the firm is low, highlighting the value of employee development within the firm. Due to regulations from the workers council, we were not able to combine age and tenure with the results from the post-experimental survey. Instead, we include categorical information on age and tenure from a different survey where no employee data was matched with survey responses. We did not collect information on employee or supervisor gender as the share of female employees in the firm is smaller than 10% and therefore not suitable for heterogeneity analysis.

Insert Table 1 about here

Main Results

We organize the results around the two alternative management-control designs embedded in the digital learning platform. The complement treatment was designed to increase supervisor attention towards employee development. Supervisors received information about whether development meetings had taken place and were prompted to conduct these conversations. This treatment therefore tests whether a technology feature that directs attention toward an underweighted development task increases supervisors' investment in employee development. We expected this treatment to increase the number of training sessions (H1a), the frequency of development interactions (H1b), and, downstream, employee knowledge (H1c). The substitute treatment was designed to remove the coordination step that ordinarily precedes a training, by allowing supervisors to act directly on employee-identified training needs through the platform without first holding a development meeting. This treatment therefore tests whether technology can substitute for direct supervisor engagement in the development process. We expected this treatment to increase the number of training sessions (H2a) and employee knowledge (H2c), but to reduce development interactions because the platform allows supervisors to bypass the meeting-based coordination step (H2b). We first test these hypotheses using the three most proximate outcomes of the development process: the number of training sessions assigned during the treatment period, the frequency of supervisor-employee development interactions, and employee knowledge after the treatment period. We further examine whether the two treatments differ in their downstream effects on job satisfaction, turnover intentions, perceived supervisor support, and operational team performance.

Results from OLS regressions (Table 2, Column 1) show that the complement treatment leads to approximately 15% more training sessions in comparison to the mean in the control group ($p = 0.089$). To investigate the robustness of this finding, we re-estimate the same regression including control variables, most importantly the number of training sessions prior to the treatments, and including both control variables and region fixed effects, which do not change our results (Table 5). In line with Hypothesis 1a, we conclude that the complement treatment increases the number of trainings scheduled by the supervisor. For the effect of the substitute treatment, we find no significant effect on the number of trainings ($p = 0.266$). If anything, the treatment led to a reduction in trainings, as evident by then negative estimate. As a result, we find no support for Hypothesis 2a.

To test the second set of pre-registered hypotheses, we regress supervisor interactions as measured by the reported number of development meetings on the treatment indicators. Results from the OLS regression (Table 2, Column 2) show that only the complement treatment significantly increases supervisor attention in form of development meetings by 13.57% ($p = 0.087$) while the substitute treatment has no significant effect ($p = 0.218$). As a result, hypothesis H1b, which stated that the complement treatment increases development interactions, is supported, while hypothesis H2b, which stated that the substitute treatment decreases development interactions, is not supported. To substantiate this result, Table 7 shows that the average time scheduled for meetings increases only in the complement treatment.

Testing the final set of pre-registered hypotheses, we regress the standardized ratio of currently answered questions in the knowledge tests after the treatment has period has ended on the treatment indicator. As column 3 of Table 2 shows, we find a positive effect on employee knowledge for both the complement treatment ($b = 0.101$, $p = 0.031$), as well as the substitute treatment ($b = 0.111$, $p = 0.001$). In Table 6, we also investigate the knowledge levels during the

treatment, however we find no significant effect for this time period. One potential reason could be that most trainings are scheduled close to or after the end of the treatment period, which is why knowledge gains might be realized at a later time point. More importantly, this lagged response further shows that knowledge gains are unlikely to be driven entirely by motivational forces resulting from increased supervision meetings or expectations about improved process management. Our findings therefore support hypotheses H1c and H2c.

Insert Table 2 about here

To assess the downstream effectiveness of the two treatments on broader employee development goals, we investigate treatment effects on employee job satisfaction, turnover intention, and perceived supervisor support. First, we regress job satisfaction, turnover intention, and perceived supervisor support on the treatment indicators. Column 1 of Table 3 shows that the complement treatment increases job satisfaction by 17.2% of an SD ($p = 0.026$), while the substitute treatment has no effect ($b = 0.027$, $p = 0.738$). In a similar vein, the complement treatment has a significant negative effect on turnover intentions ($b = -0.153$, $p = 0.073$), while the substitute treatment has no significant effect on turnover intentions ($b = -0.08$, $p = 0.288$). Regarding supervisor support, the complement treatment shows a positive effect ($b = 0.174$, $p = 0.081$), while the substitute treatment has no effect ($b = 0.116$, $p = 0.241$).

Insert Table 3 about here

Finally, we examine whether the treatments translate into improvements in actual operational performance, measured as the average time required to complete different categories of work

assignments. To do so, we estimate difference-in-differences models using assignment-level data while controlling for team and region-by-week fixed effects, and additionally include task fixed effects in the pooled specification. As shown in Table 4, the complement treatment significantly reduces the average completion time across all assignments by 0.447 minutes ($p = 0.048$), corresponding to roughly a 3% improvement relative to the pre-treatment mean of 15.37 minutes. The substitute treatment also yields a negative coefficient but the effect is somewhat smaller (-0.401 minutes, $p = 0.077$). Decomposing the treatment effect by task category reveals that these gains are driven almost entirely by troubleshooting activities, which are the most technically demanding and time-intensive assignments. The complement treatment reduces troubleshooting time by 1.308 minutes per assignment (Column 3, $p = 0.031$), or a roughly 4% improvement, whereas the time reduction for the substitute treatment is – at least descriptively – distributed across all tasks. Taken together, these findings suggest that increasing supervisor attention to employee development primarily improves performance in tasks where accumulated technical knowledge and skill is most valuable, while simpler or more standardized activities remain largely unaffected.

 Insert Table 4 about here

Finally, we examine whether the treatments generate unintended quality trade-offs. As shown in Table 8, neither the complement nor the substitute treatment significantly affects the share of unresolved assignments or customer service ratings. These findings indicate that directing supervisor attention toward employee development improves skill-intensive operational performance without reducing other dimensions of service quality.

 Insert Table 8 about here

Can technology substitute supervisor engagement?

Taken together, the results allow us to address the question that motivates the study: whether the platform can substitute for the supervisor's engagement in producing development outcomes. The pattern across outcomes is asymmetric. On the proximate output of the development process, employee knowledge, the two treatments are comparable as both raise post-treatment knowledge. On every measured outcome that involves the supervisor's engagement, however, only the complement treatment moves the outcome. The complement treatment increases the number of training sessions (approximately 15%, $p = 0.089$) and the frequency of development meetings (approximately 14%, $p = 0.087$), and increases the time supervisors spend in development interactions, an increase concentrated in informal rather than formal meetings ($b = 0.157$, $p < 0.01$; Table 7). The substitute treatment moves none of these: it does not increase training sessions ($p = 0.266$), development meetings ($p = 0.218$), or time spent in meetings of any kind.

The same asymmetry appears in the broader outcomes the firm values. The complement treatment increases job satisfaction ($p = 0.026$), increases perceived supervisor support ($p = 0.081$), and reduces turnover intentions ($p = 0.073$), while the substitute treatment has no detectable effect on any of these ($p = 0.738$, 0.241 , and 0.288 , respectively). In operational performance, the complement treatment reduces completion time by about 3% (0.447 minutes, $p = 0.048$), with the reduction concentrated in troubleshooting (1.308 minutes, $p = 0.031$), the most skill-intensive task. The substitute treatment also reduces overall completion time, but the effect is spread across tasks rather than concentrated in skill-intensive work, so it shows less of a skill-driven improvement.

This pattern speaks directly to the substitution question. The platform feature in the substitute treatment reproduced the proximate skill outcome, employee knowledge, but did not reproduce the broader outcomes that the complement treatment produced and that depend on the

supervisor's engagement with the employee: perceived support, satisfaction, retention, and the skill-concentrated operational gain. In our setting, then, technology could substitute for the supervisor in delivering the measurable skill gain, but not in delivering the relational and operational benefits that flow through supervisor engagement.

A notable feature of the substitute treatment's knowledge effect is that it arose without any increase in the measured behaviors that would produce it. The number of training sessions did not rise, the number of development meetings did not rise, and the time spent in meetings did not rise. The channel through which the substitute treatment raised knowledge is therefore not directly visible in our data, and we can only speculate about it. One possibility is that allowing employees' self-identified development needs to reach action without the intermediating meeting improved the match between the trainings employees received and their actual skill gaps, raising the knowledge produced per training even without an increase in the number of training sessions. A second possibility is that the treatment shifted the timing of training earlier, by letting supervisors act on identified needs without waiting for the next scheduled meeting, leaving more time for skill gains to accumulate before they were measured. We cannot distinguish between these explanations with our data. What the result establishes, regardless of the channel, is that the knowledge gain in the substitute treatment did not operate through increased supervisor engagement, since none of the engagement margins moved. The further finding that neither treatment raised knowledge during the treatment period, with gains emerging only afterward (Table 6), is consistent with this reading: the knowledge gains reflect lagged skill acquisition rather than a contemporaneous motivational response to supervisor attention.

Further, the quality trade-off results bear on a central concern in the multitasking setting that motivates our study. If directing supervisors' attention toward the unmeasured development task pulls effort away from the operational tasks the firm measures and rewards, then the

treatments that increase development activity should degrade operational performance (Holmstrom and Milgrom 1991; Feltham and Xie 1994). We find no evidence of such crowding out on the dimensions the firm measures. The complement treatment, which increased training, development meetings, and the time supervisors spent in development interactions, did not raise the share of unresolved assignments or lower customer service ratings (Table 8), and it reduced rather than increased job-completion time (Table 4). Directing supervisor attention toward development therefore did not come at the expense of the measured operational tasks, and in the case of the complement treatment it improved the most skill-intensive one. We cannot rule out crowding out on dimensions the firm does not measure, but on the operational outcomes the firm tracks and values, the reallocation of attention toward development was not offset by deterioration elsewhere.

5. CONCLUSION

In this study, it was our goal to support employee development in the workplace to test simple treatment by substituting or complementing supervisor contact using new features on a digital learning platform. We find that the complement treatment has positive consequences for the workforce, confirming most of our pre-registered hypotheses. As hypothesized, we document an increase in trainings scheduled by the supervisor, as well as an increase in supervisor contact for the complement treatment. Regarding the downstream effects of the treatments, we find that both treatments lead to an increase in employee knowledge, however only the complement treatment has a positive effect on job satisfaction, reduced turnover intentions, and perceived supervisory support, as well as troubleshooting performance, as measured by the average time per work assignment.

In our setting, conducting the experiment has convinced the workers council of the firm to release all function associated with the complement treatment to all employees. A back-of-the-envelope calculation shows that even small interventions can produce potentially large returns.³ Normalizing the costs of implementing all changes associated with the experiment to one and calculating the estimated annual savings in personnel costs from the complement treatment, predict an ROI of 807%. By the nature of our field experiment, our evidence can be generalized to many work contexts and supervisors can closely follow our approach if they seek to implement the complement treatment. While we provide causal evidence from a large real-world sample, our results might be less applicable to workplace environments with increased supervisor attention. The baseline interaction frequency regarding skill development is low in our context ($M = 2.13$ over a period of 18 months), making the potential for an effective treatment more likely. Future research could look at organizational interventions that can support training and development in more psychologically close environments. Further, while we provide evidence on the effectiveness of our treatments, we remain agnostic about how our treatments affect the decision-making process on the supervisory level and their perception of the learning platform. Finally, our experimental population is older, more male-dominated and exhibits substantially low turnover in comparison to the average firm. Because of this, we are unable to investigate treatment effects on realized voluntary turnover and heterogeneous treatment effects with respect to employee or supervisor gender, which can be important in the context of supervisor attention.

³ An important caveat in this calculation is that training costs are not considered. Unfortunately, we currently do not have access on the realized hours of training. Calculating the ROI under the assumption that each training takes half a day per employee and the effect of training does not carry over to the next year, the estimated ROI is 527%.

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Figure 1

Development Process of the Firm and Experimental Conditions

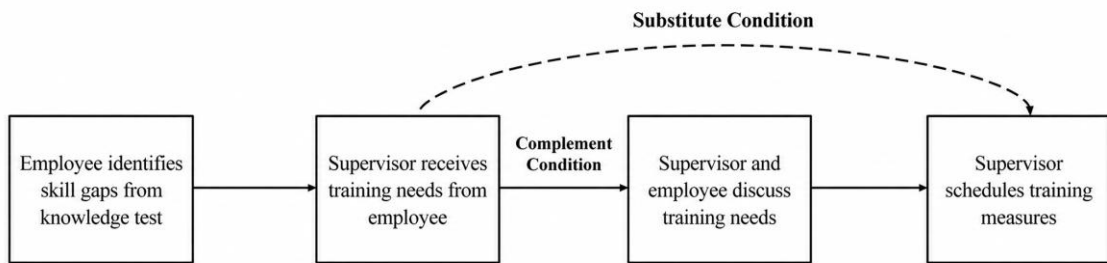


Table 1
Descriptive Statistics and Randomization Checks

Variable	Full Sample	Control Treatment	Complement Treatment	Substitute Treatment	F-statistics
Trainings	5.45 (6.11)	5.64 (6.49)	5.82 (6.47)	4.87 (5.38)	0.42 ($p = 0.65$)
Knowledge	0.00 (0.28)	0.01 (0.25)	0.00 (0.28)	0.00 (0.32)	0.02 ($p = 0.97$)
Development Meetings	2.13 (0.78)	2.08 (0.88)	2.09 (0.71)	2.23 (0.74)	0.59 ($p = 0.55$)
Perceived Supervisor Support	5.26 (0.81)	5.14 (0.90)	5.26 (0.78)	5.39 (0.75)	1.16 ($p = 0.31$)
Team Size	20.12 (2.59)	20.37 (2.69)	20.00 (2.16)	20.02 (2.91)	0.40 ($p = 0.66$)
Team Performance	34.44 (24.17)	33.99 (23.81)	34.40 (24.51)	34.90 (24.18)	0.02 ($p = 0.98$)
Age Group in %					
18-29 years	14.24				
30-49 years	25.15				
>50 years	60.59				
Tenure in %					
0-4 years	4.25				
5-10 years	8.64				
>10 years	87.10				
<i>N</i>	3,644	1,247	1,190	1,207	
Teams	187	62	63	62	

Note. Table shows means and standard deviations of key variables for the full sample and each treatment conditions separately. The F-tests as to whether the treatment conditions exhibit significant differences prior to the intervention are reported in the column on the far right of the table. Age and tenure statistics are reported from a separate survey with $N = 1,346$ employees. For all other variables we use data from all 187 teams and 3,644 employees, respectively.

Table 2
Treatment Effects on Trainings, Supervisor Attention, and Employee Knowledge

	<i>Dependent variable:</i>		
	Number of Trainings	Supervisor Attention	Employee Knowledge
	(1)	(2)	(3)
<i>Complement Treatment</i>	6.364* (8.435)	1.428* (0.779)	0.101** (0.045)
<i>Substitute Treatment</i>	-2.572 (2.314)	-0.051 (0.744)	0.111*** (0.042)
P-value Comp. vs. Sub.	$p = 0.014$	$p = 0.055$	$p = 0.507$
Controls	Yes	Yes	Yes
Observations	171	1,286	2,393
Teams (Clusters)	171	187	182
R ²	0.043	0.200	0.323

Note. Column (1) shows a result from OLS regressions on the number of trainings assigned by the supervisor in each experimental treatment during the treatment period. Robust standard errors are shown in parentheses, Column (2) and (3) show results from OLS regressions estimating the average treatment effect on supervisor contact (2), defined as the reported number of employee development meetings, and employee knowledge (3), defined as the scaled ratio of correctly answered questions on all post-treatment knowledge tests in the post-treatment period (March 2023 – December 2023). Controls across all models include pre-treatment values and the number of skills per team. Standard errors are clustered on the team level. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Table 3

Treatment Effects on Job Satisfaction, Turnover Intention, and Supervisor Support

	<i>Dependent variable:</i>		
	Job Satisfaction	Turnover Intention	Supervisor Support
	(1)	(2)	(3)
<i>Complement Treatment</i>	0.171** (0.076)	-0.153* (0.085)	0.174* (0.099)
<i>Substitute Treatment</i>	0.027 (0.081)	-0.089 (0.083)	0.116 (0.099)
Constant	-0.065 (0.052)	0.079 (0.063)	-0.095 (0.076)
Wald Test	$p = 0.062$	$p = 0.196$	$p = 0.211$
Observations	1,165	1,239	1,271
Teams	187	187	187
R ²	0.006	0.004	0.005

Note. Table shows a series of OLS regressions of employee survey responses assessing (1) job satisfaction, (2) turnover intention, and (3) perceived supervisor support. All variables are standardized, and the table reports estimates as changes in percentage of a standard deviation. Standard errors are clustered on the team level and included in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Table 4
Treatment Effects on Assignment Performance

	<i>Dependent variable:</i>				
	All Tasks (1)	Customer Service (2)	Trouble- shooting (3)	Quality Control (4)	Docu- mentation (5)
<i>Complement Treatment</i>	-0.447** (0.022)	0.197 (0.362)	-1.308** (0.602)	0.178 (0.447)	-0.115 (0.114)
<i>Substitute Treatment</i>	-0.401* (0.022)	0.166 (0.314)	-0.539 (0.624)	-0.331 (0.363)	-0.175 (0.119)
DV mean	15.366	7.255	33.994	10.584	4.924
Clusters (Teams)	187	187	187	187	187
Observations	223,047	50,721	63,777	51,773	56,744
Team FE	Yes	Yes	Yes	Yes	Yes
Region-Week FE	Yes	Yes	Yes	Yes	Yes
Task FE	Yes	No	No	No	No
R ²	0.391	0.105	0.219	0.116	0.267
Within R ²	0.001	0.001	0.001	0.001	0.001

Note. Table shows estimates from difference-in-differences regressions of the average treatment effects as measured by the total amount of time spent on a given activity by all members of a team. Column 1 shows results across the four core tasks while columns 2-5 show separate regressions per task. All regressions include team fixed effects and regional-specific time trends to account for local variation due to regional holidays. Standard errors are clustered on the team level and included in parentheses. * $p < 0.05$, ** $p < 0.01$

Table 5
Robustness Checks: Treatment Effect on Team-Level Trainings

	<i>Dependent variable: Number of Trainings</i>		
	(1)	(2)	(3)
<i>Complement Treatment</i>	0.457* (0.269)	0.447* (0.266)	0.388* (0.218)
<i>Substitute Treatment</i>	-0.326 (0.258)	-0.301 (0.266)	-0.281 (0.256)
Constant	2.324*** (0.186)	2.109*** (0.211)	
Wald Test	$p = 0.014$	$p = 0.015$	$p = 0.014$
Control	No	Yes	Yes
Fixed Effects	No	No	Yes
Observations	171	171	171
R ²	0.043	0.075	0.438
Adjusted R ²	0.052	0.077	0.387

Note. Table shows a series of Poisson regressions predicting the count of assigned trainings by the supervisor in each experimental condition. We exclude teams for which no training data was available for the entire post- and pre-treatment period. Control variables include the count of assigned trainings prior to the intervention and team size. Fixed Effects account for the department structure of the firm. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Table 6
Treatment Effects on Employee Knowledge

	<i>Dependent variable: Employee Knowledge</i>	
	Treatment Period	Post-Treatment Period
	(1)	(2)
<i>Complement Treatment</i>	-0.004 (0.044)	0.101** (0.045)
<i>Substitute Treatment</i>	-0.006 (0.042)	0.111*** (0.042)
Wald Test	$p = 0.926$	$p = 0.729$
Controls	Yes	Yes
Employees	2,796	2,393
Teams	182	182
R ²	0.323	0.265

Note. Table shows estimates from regression of the standardized knowledge test score across all knowledge tests conducted during the experimental periods (Column 1) and all knowledge tests over the period of one year after the experiment has been discontinued (Column 2). Standard errors are clustered on the team level and included in parentheses. All analyses include controls for the average knowledge test results prior to the treatment. Standard errors are clustered on the team level and included in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Table 7
Treatment Effects on the Scheduled Time for Development Activities

	<i>Dependent variable:</i>		
	All Meetings (1)	Formal Meetings (2)	Informal Meetings (3)
<i>Complement Treatment</i>	0.124** (0.056)	0.007 (0.028)	0.157** (0.063)
<i>Substitute Treatment</i>	0.045 (0.046)	-0.018 (0.032)	0.063 (0.056)
Observations	46,418	7,188	38,631
Clusters (Teams)	187	187	187
Team FE	Yes	Yes	Yes
Region-Week FE	Yes	Yes	Yes
Activity FE	Yes	No	No
R ²	0.169	0.415	0.179
Within R ²	0.000	0.000	0.000

Note. Table shows estimates from difference-in-differences regressions of the average treatment effects as measured by total amount of time spent on a given activity by all members of a team. Time is measured in full-time-equivalents (FTE). Standard errors are clustered on the team level and included in parentheses. All regressions include team-specific, and week- fixed effects. In Column 1, the week-fixed effect is interacted with an activity-specific fixed effect to account for seasonal variation in the scheduling of specific activities. + p < 0.1; * p < 0.05; ** p < 0.01; *** p < 0.001

Table 8
Treatment Effects on Work Quality

	<i>Dependent variable:</i>	
	Unresolved Assignments (1)	Customer Service Rating (2)
<i>Complement Treatment</i>	-0.000 (0.001)	-0.013 (0.011)
<i>Substitute Treatment</i>	-0.001 (0.001)	-0.013 (0.011)
Observations	13,700	10,651
Teams	182	182
Team FE	Yes	Yes
Region-Week FE	Yes	Yes
R ²	0.032	0.042

Note. Table shows estimates from difference-in-differences regressions of the average treatment effects on two measures of work quality. Column (1) reports results for the share of unresolved assignments, defined as customer cases that required an additional follow-up visit, while column (2) reports the weekly average customer service rating on a scale from 1-5 obtained from post-service customer feedback surveys. All regressions include team fixed effects and region-specific week fixed effects to account for local seasonal variation, such as regional holidays. Standard errors are clustered at the team level and reported in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$